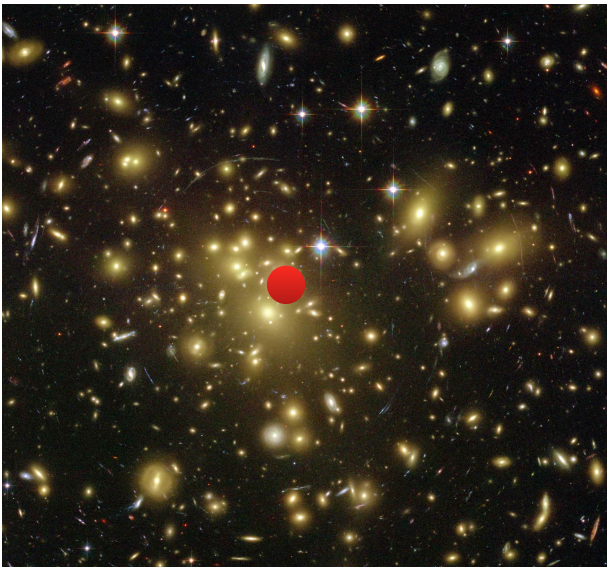


Jigsaw Expert Activity

- Step 1:
 - You now have one image with a caption. Take 1-2 minutes to read the caption, and try to understand the figure.
 - Find 1-2 people near you with the same picture - you should be able to do this without getting up
 - Discuss the card together, and see if you have a guess about what it is showing.
- Step 2:
 - After five or so minutes, I will make an announcement. At that time, split up, and form new groups of three people, where each person has a different card. You should be able to do this without traveling too far.
 - Go around your group, share your expert knowledge of the topic of your card, and take questions from your group-mates. Try to figure out what these images together are showing you.
- Step 3:
 - We'll have 2-3 groups volunteer to share what they have discovered.
 - We will collect your images for use in the next class.



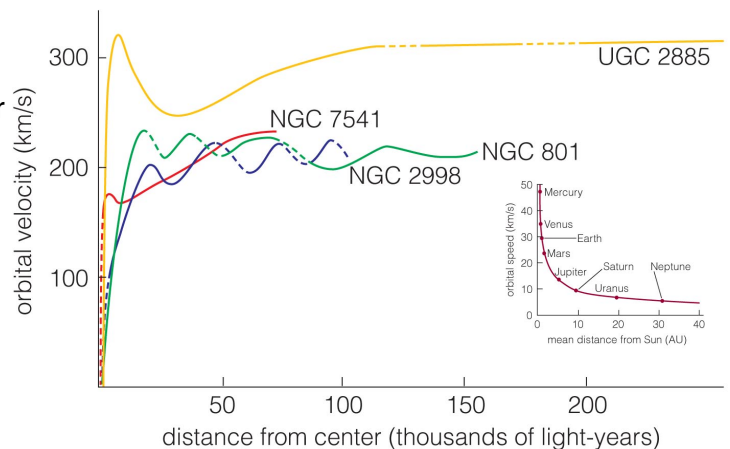
Speed of objects in galaxy clusters - All of the galaxies in this image exhibit **random orbits** around the center of mass, shown by the red dot. Estimating the mass of stars by adding up all of the light seen, Fritz Zwicky in 1933 estimated that the **velocity dispersion** (the spread of line-of-sight velocities) of galaxies in the Coma cluster (which is about 300 million light years from the Milky Way) should be around 80 km/s. However, when he took spectra of many galaxies, he found that the velocity dispersion was more like 1500 - 2000 km/s.

What does this mean, and why do you think it is?

Rotation curves of spiral galaxies - This figure shows the orbital velocity as a function of the distance from the center of a galaxy for several spiral galaxies. Remember that in our Solar System (shown in the lower-right inset), the orbital speed falls with distance from the Sun.

What does this say about the mass distribution of our Solar System?

If we look at spiral galaxies, most of their light does appear concentrated towards the center. *What does the flat rotation curve mean about the mass distribution?*



Colliding galaxy clusters - This image shows the collision of two galaxy clusters, known as the Bullet Cluster. The image showing the galaxies was taken by the Hubble, and shows where the galaxies (and thus the stars) are after the collision. These clusters have already passed through each other, and since the distances between galaxies are large, they just kept going (note the two clustered regions of galaxies in the image). However, clusters have *lots* of hot gas, about 10X as much mass in gas as in stars, and when clusters collide, this gas slams together, and screeches to a halt. The pink region denotes this hot gas, showing that it got stopped, while the galaxies kept going. Finally, the blue color denotes where most of the mass is, determined by gravitational lensing. Remember that in galaxy clusters, the mass should be dominated by gas and not stars.



Why do you think most of the mass is still located with the galaxies?